Macroeconomic Regimes and Apple's Equity Performance

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DATA 200: Data Systems for Data Analytics

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INTRODUCTION

Financial markets do not operate in a vacuum outside of general macroeconomic conditions. Inflation dynamics, monetary policy, recessions, and real economic growth all determine risk premia, discount rates, and investor perceptions, which in turn influence asset prices. However, the degree to which macroeconomic conditions influence the long-term performance of individual firms remains a subject of debate. This project extends my Project 1 analysis of Apple Inc. (AAPL) by explicitly incorporating macroeconomic time series from the Federal Reserve Economic Data (FRED) API. The objective is to determine whether Apple's equity performance is essentially a function of macroeconomic regimes or firm-specific fundamentals that dominate over long horizons.

The project proceeds in four stages. First, I construct a reusable Python function to retrieve macroeconomic data directly from the FRED API. Next, I transform the raw API output into tidy, mergeable tabular datasets prepared for analysis. I then visualize key macroeconomic relationships to establish the context of the macroeconomy, and subsequently combine these macro indicators with Apple's stock price data to analyze how macroeconomic regimes interact with Apple's performance and how Apple compares to the broader market. Together, these steps provide a data-driven framework for understanding the role of macroeconomic forces in equity valuation.

API STRUCTURE, PATH, AND DATA RETRIEVAL

The project retrieves macroeconomic data from the Federal Reserve Economic Data (FRED) API using the endpoint https://api.stlouisfed.org/fred/series/observations. FRED structures economic indicators as a set of standardized time series, where each series has a unique series_id (e.g., CPIAUCSL for consumer price inflation, USREC for recession indicators, FEDFUNDS for

the federal funds rate). This structure enables the same API path to be reused across data sets and customization be handled via query parameters, rather than changes to the URL itself.

The custom function exploits this design by holding the endpoint constant and varying parameters such as series_id, observation_start, observation_end, frequency, and aggregation_method. These parameters significantly alter the data retrieved, enabling the same function to extract inflation, GDP, interest rates, or market indices across different time horizons or temporal resolutions. By using an API key to authenticate requests made to FRED and relying on HTTP statuses to verify data validity, the function returns observations in a Pandas DataFrame. Because this function relies on FRED's standardized schema, it offers a flexible and reproducible mechanism by which macroeconomic data can be acquired for downstream analysis and visualization.

ENSURING TIDY TABULAR DATA

Raw FRED data are returned as JSON objects with date and value fields stored as strings. To ensure tidy data principles, each dataset is transformed so that every row represents a single time period, and each column represents a single variable. Dates are converted to proper datetime objects, values are coerced to numeric types with invalid entries removed, and columns are renamed to descriptive variable names. The data are then sorted chronologically and indexed consistently. This process ensures that all macroeconomic series share a common structure and can be merged reliably with each other and with Apple’s stock price data using a unified date key.

MACROECONOMIC VISUALIZATIONS AND INTERPRETATION

Inflation and Recessions

The first visualization plots year-over-year CPI inflation with recession periods shaded. The chart demonstrates that inflation behavior is regime-dependent rather than uniformly linked to recessions. The early extreme high inflation rates in the early 1980s corresponded with successive recessions, reflecting the Federal Reserve's aggressive monetary tightening under Chairman Paul Volcker to break entrenched inflation expectations (Goodfriend & King, 2005). In contrast, the 2008–2009 recession aligns with sharp disinflation and temporary deflation, driven by a collapse in aggregate demand following the financial crisis. The post-2020 episode stands apart: instead of occurring during a recession, inflation spiked during the recovery phase, reflecting supply-chain disruptions, fiscal stimulus, and rapid reopening dynamics (Blanchard, 2022).

The key insight is that recession indicators alone do not explain inflation outcomes. Instead, the nature of the underlying shock, such as supply-side versus demand-side, and the credibility of monetary policy play a decisive role. For the investor, this is a distinction with a difference because inflation itself serves as a harbinger of interest rates and discount factors that will influence equity valuations, even in the absence of contraction.

GDP Growth, Inflation, and Monetary Policy

The second visualization combines real GDP growth, CPI inflation, and the federal funds rate into a single macroeconomic dashboard. The chart highlights a consistent sequencing pattern: monetary policy is responding to inflation and growth conditions rather than leading them. During the Global Financial Crisis, GDP growth turned sharply negative, inflation fell, and the Federal Reserve lowered rates toward the zero lower bound. Conversely, in the post-pandemic period, inflation accelerated during economic recovery; beginning in 2022, one of the fastest tightening cycles in decades is underway.

This pattern aligns with standard monetary policy reaction functions, such as the Taylor rule, which emphasize stabilizing inflation and output gaps (Taylor, 1993). For equity markets, the implication is that macroeconomic conditions are transmitted into asset prices primarily through discount-rate channels. Growth may remain positive, but rising rates can still pressure equity valuations. This mechanism is particularly relevant for high-duration equities, whose valuations are more sensitive to changes in expected discount rates.

APPLE, MACROECONOMICS, AND MARKET COMPARISON

Apple Stock, Recessions, and Bear Markets

At this stage, macroeconomic indicators are merged with Apple's stock price data to examine how economy-wide conditions interact with firm-level performance. The first combined visualization overlays Apple's adjusted stock price with recession periods, GDP growth, and market-wide bear regimes. The chart reveals a pronounced asymmetry. Recessions consistently coincide with heightened volatility and drawdowns, but they do not dictate Apple's long-run trajectory. Apple experiences sharp declines during macro shocks, most notably during the 2008–2009 crisis and the 2020 pandemic, but recoveries are rapid and often stronger than the broader economy.

This pattern suggests that macroeconomic stress primarily affects Apple through risk repricing and liquidity conditions rather than through permanent damage to earnings capacity. GDP growth often recovers after equity prices have already rebounded, underscoring that financial markets are forward-looking. For decision-making, the implication is that recession signals are more informative for short-term risk management than for long-term valuation of firms with durable competitive advantages.

Apple versus the S&P 500

The final visualization compares Apple's performance to the S&P 500 using indexed levels and cumulative excess returns. Indexing isolates relative wealth creation rather than absolute price movements. The results show persistent and substantial outperformance by Apple over the sample period. The cumulative excess return remains positive and increases over time, indicating that Apple's returns cannot be explained solely by market beta.

This sustained outperformance reflects firm-specific factors such as innovation, ecosystem lock-in, pricing power, and scale economies. However, the analysis also highlights a forward-looking constraint: persistent outperformance embeds high expectations into valuations. As expectations rise, future returns become more sensitive to macroeconomic conditions that affect discount rates and earnings surprises. Thus, while Apple's historical performance demonstrates strong fundamentals, macroeconomic regimes still matter for timing and risk assessment.

REASSESSMENT, NEW QUESTIONS, AND NEXT STEPS

Combining this with the macroeconomic data reinforces my finding from Project 1: it is firm-specific fundamentals that drive Apple's long-run performance, while the impact of macroeconomic regimes is primarily reflected in short-term volatility and valuation pressure. Among these, inflation and monetary policy prove to be more immediate channels than recession indicators.

Future research, incorporating additional data sources such as news sentiment, earnings call tone, or measures of market uncertainty, can help capture non-macroeconomic drivers of price movements. A natural next step would be to build a predictive model that combines macro variables with firm-level indicators and sentiment measures to forecast returns or volatility regimes. Such a model could improve decision-making by distinguishing between macro-driven drawdowns and changes driven by shifts in firm fundamentals.

References

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